Macro drivers of the yield curve

Stage 1 | Research design, variable selection and tested starter code

Prepared 14 October 2026 | Retrospective report on the saved project package

Decision: establish an auditable benchmark first

Stage 1 selected a small quarterly macro-augmented autoregression (ARX) for 2-, 5- and 10-year Treasury yield changes. The baseline uses revisions to expected real GDP growth and expected CPI inflation. Borrowing, Fed holdings, oil, unemployment and policy expectations are evaluated in separate extensions.

The aim is to test whether information available at quarter-end improves next-quarter yield forecasts over no-change and autoregressive benchmarks. The design is motivated by macro-finance literature, but this exact specification is a project choice rather than a universally optimal model.

Delivered in Stage 1	Evidence status
Economic channels and bibliography	Research design and source review.
Real Treasury reference curve	9 October 2026: 2Y 4.80%, 5Y 5.02%, 10Y 5.24%; 2s10s +44 bp.
Python estimator and SQL schema	Working starter implementation and vintage-aware data contract.
Five integrity tests	Passed in the saved validation record.
Baseline estimation exercise	120 synthetic quarters only; no real macro coefficient or forecast evidence.

Stage 1 is now a historical design milestone. The subsequently completed Stage 2 report supplies the real-data estimation and benchmark results. Synthetic Stage 1 scores must never be mixed with Stage 2 empirical results.

1. Economic channels and selected variables

A nominal long yield can be viewed approximately as expected average future short rates plus a term premium. Growth and inflation outlooks influence the anticipated policy path. Borrowing and central-bank holdings influence the amount and composition of interest-rate risk investors absorb. The small regression does not separately identify these channels.

Topic	Chosen measurement / role	Key qualification
Growth	Same-target SPF next-quarter real GDP growth revision; baseline	Derived from median levels, not median individual growth forecasts.
Inflation	Same-target SPF next-quarter CPI inflation revision; baseline	Survey revision is not an identified announcement shock.
Borrowing	Change in debt held by the public / lagged annual GDP; supply extension	Stock change is not exact issuance or fiscal news.
Central bank	Change in Fed Treasury holdings / lagged annual GDP; supply extension	Fed Treasury portfolio only; global balance sheets deferred.
Oil	Log change in quarterly-average Brent; separate extension	Demand and supply shocks can have different effects.
Unemployment	Change in latest available three-month mean; substitute for growth	Shares cyclical information with GDP.
Policy outlook	Same-target SPF Treasury-bill forecast revision; robustness check	May absorb part of the macro transmission channel.

The model uses revisions rather than simply the level of inflation or GDP growth because previously anticipated conditions may already be priced. Even survey revisions may be priced before quarter-end, so incremental forecasting value can be weak or zero.

2. Specification, timing and alternatives

For each maturity: next-quarter yield change = intercept + persistence x current-quarter yield change + growth coefficient x growth revision + inflation coefficient x inflation revision + error. Yield changes are measured in basis points. Estimate by ordinary least squares with four-lag HAC coefficient covariance.

The 2s10s prediction equals the predicted 10Y change minus the predicted 2Y change. A positive difference means predicted steepening. The 5Y equation adds information about the belly. Three constant-maturity points do not constitute a continuous zero-coupon or arbitrage-free yield curve.

GDP construction compares annualized growth for the same target quarter using within-vintage level ratios: 100 x [(RGDP3/RGDP2)^4 - 1] in the current survey, minus 100 x [(RGDP4/RGDP3)^4 - 1] in the preceding survey. Inflation uses CPI3 now minus CPI4 previously. Within-vintage ratios avoid comparing incompatible GDP base levels.

Borrowing and Fed inputs are first normalized to billions. Their quarterly stock changes are divided by lagged annualized nominal GDP and multiplied by 100; the flows themselves are not annualized. Use only endpoints and denominators observable by the origin, and retain reference periods and input ages.

Alternative	Stage 1 decision
Macro factor / Nelson-Siegel or VAR	Potential next description of level, slope and curvature with feedback.
No-arbitrage macro-finance model	Defer: richer yield coverage, estimation and pricing restrictions required.
Local projections with identified shocks	Appropriate for causal questions only with credible shock identification.
Black-box machine learning	Defer until a simple benchmark, sufficient data and credible validation exist.

Predicted signs are hypotheses, not imposed restrictions. Stronger growth or inflation may lift yields, but curve shape depends on which maturities respond. Oil is ambiguous; unemployment often lowers policy expectations; Fed purchases may lower long yields. None implies a universal basis-point multiplier.

3. Validation plan, status and references

The original plan targets quarterly data from 2003 onward, subject to overlap, and at least 60 usable training quarters in expanding-window estimation. It compares RMSE and MAE with no-change and AR-only forecasts on identical dates. Preserve the calendar grid before creating lags; never replace missing quarters with compressed time steps.

Store reference dates, release dates, vintage dates, units and sources. No training target may lie beyond the forecast origin. Inspect coefficient uncertainty, residual correlation, collinearity, persistence and sensitivity to pandemic/lower-bound periods. Coefficient confidence intervals are not forecast prediction intervals.

Recorded Stage 1 checks: future-feature rejection; quarterly calendar continuity; lag/target alignment; GDP rebasing and balance scaling; SQL exclusion of unavailable vintages. The synthetic baseline ran on 120 quarters with 60 initial training rows. This verifies implementation behavior, not real-market forecasting accuracy.

To reproduce the historical starter, extract Macro_Yield_Model_Stage1.zip, install requirements.txt, then run python model.py --demo. Demonstration outputs are explicitly labeled SYNTHETIC_DEMO. Use the Stage 2 package for real macro data and empirical comparisons.

Transition to Stage 2: real-data availability reduced the supply/oil sample and required a 40-row initial window for those exploratory comparisons; oil uses the previous full quarter; historical survey corrections and current yield history limit vintage purity. These implementation departures belong to Stage 2 and should not be retroactively described as completed Stage 1 work.

Report evidence: Stage 1 MODEL_DESIGN.md, SOURCES.md, VALIDATION.md, README.md and yield snapshot. Selected literature: Ang and Piazzesi, https://www.nber.org/papers/w8363 ; Diebold, Rudebusch and Aruoba, https://econ.umd.edu/publication/macroeconomy-and-yield-curve-dynamic-latent-factor-approach ; Li and Wei, https://www.federalreserve.gov/pubs/feds/2012/201237/ ; Orphanides and Wei, https://www.federalreserve.gov/pubs/feds/2010/201001/index.html .

Codex-assisted development covers research synthesis, code, SQL timing controls and software verification. No language-model output enters the forecasting equation as a market signal. The later Stage 2 finding that the baseline did not beat simple benchmarks is a valid research result, not a reason to relabel synthetic tests as evidence.